Manuel Jimenez

AI ENGINEER • SENIOR BACKEND DEVELOPER

manuel28042005@gmail.com | linkedin.com/in/manueljimenezlopez | github.com/manuel-jimenez10 | manueljimenez.vercel.app | LATAM (Remote)

PROFESSIONAL SUMMARY

AI Engineer and Senior Backend Developer specialized in Python, LLM applications, multi-agent systems, and microservices architectures. Expert in building scalable APIs, process automation, and LLM integration for SaaS products. Solid experience in the Python ecosystem for high-performance deployments, complemented by a versatile proficiency in multiple languages and infrastructure tools, ensuring robust and efficient enterprise-grade systems.

CORE COMPETENCIES

Multi-Agent LLM Systems (LangGraph, CrewAI) | RAG & Vector Search (Qdrant) | Python / FastAPI | Node.js / NestJS / TypeScript | Meta WhatsApp Business API | CRM Integrations (EspoCRM) | Redis & Celery Async Workers | CI/CD & Docker

WORK EXPERIENCE

SellEasy

Jan 2026 - Present

AI Engineer • LLM Systems & Multi-Agent Specialist

Medellin, Colombia (Remote)

- Architected conversational AI multi-agent workflows (LangGraph, CrewAI, RAG) for full-cycle e-commerce sales and customer support, resolving user queries autonomously and cutting response latency by **32%**.
- Built autonomous ad-ops AI agents integrated across Meta Ads, TikTok Ads, and Google Ads APIs, automating campaign orchestration and budget tracking to reduce marketing management time by **23%**.
- Led technical integration protocols and compliance to achieve official Meta Certified Tech Provider accreditation, centralizing multi-tenant merchant onboarding and WhatsApp API pipelines.
- Designed semantic search and retrieval systems using Qdrant vector database and embeddings, optimizing token utilization, prompt caching, and cost-per-inference with Helicone observability.

Netcare IT

Jun 2024 - Sep 2026

Senior Backend Engineer • AI Integrations & API Specialist

Cabo San Lucas, Mexico (Remote)

- Architected and deployed an asynchronous Meta WhatsApp Business Cloud API integration directly into the enterprise CRM (EspoCRM), automating quote workflows and pre-invoicing pipelines to cut administrative workload by **21%**.
- Built automated quote follow-up workflows inside EspoCRM, using cron jobs and webhooks to send WhatsApp notifications through Twilio.
- Built an event-driven EspoCRM → Mailchimp contact sync: every contact created, updated, or deleted in the CRM is propagated to Mailchimp automatically (full CRUD sync).
- Engineered high-concurrency RESTful API endpoints and WebSocket channels for centralized bidirectional omnichannel messaging, boosting commercial team sales output by **22%**.
- Implemented end-to-end CI/CD automated deployment workflows and microservices orchestration on Render and cloud infrastructure, eliminating manual deployment friction and standardizing zero-downtime releases.
- Integrated distributed caching and asynchronous background task workers using Redis and Celery to process high-volume messaging queues without system degradation.

Seagloo

Jul 2022 - Aug 2024

Backend Developer

Buenos Aires, Argentina (Remote)

- Developed an AI emergency detection engine with real-time geospatial tracking connecting into 911 dispatch networks, slashing emergency alert transmission time by **17%**.
- Integrated and evaluated large language models (LLM pipelines based on ChatGPT-4o) with prompt validation layers, boosting critical incident classification accuracy by **20%**.
- Containerized multi-service backend applications using Docker and Docker Compose, mitigating environment parity drift by **15%** across development, staging, and production environments.
- Designed event-driven messaging and microservice data streams using Node.js, Express, and PostgreSQL to ensure zero packet loss during critical notification bursts.

PROJECTS

Evento Maiawa - NFC & Voice Event Platform (built at SellEasy)

Built the software for "Lost in Paradise", an in-person event (Cartagena, Nov 2026) combining NFC passports, WhatsApp missions, voice clues played on a vintage telephone, and an NFC-locked treasure chest, with hardware by partner UINN. Designed the device API that NFC readers call on every tap (card UID + shared token → numeric action code + MP3 in base64 for the device speakers), with heartbeats for a live reader-health dashboard and fail-safe "deny" responses so a reader never hangs. Integrated ElevenLabs text-to-speech (eleven_multilingual_v2) to pre-generate personalized voice clues, tuning output to mono 32 kbps MP3 to shrink a 10-second clip from ~213 KB to ~54 KB of base64. Built WhatsApp messaging through SellEasy's bridge: Meta template management, batched mass sends, yes/no intent detection on free-text replies, per-button auto-replies, and an HMAC-SHA256-signed webhook.

TypeScript, Next.js, Supabase, ElevenLabs API, WhatsApp Business API, NFC hardware

Cobrix - AI-Powered Debt Collection via WhatsApp

Built a WhatsApp-based collections app for Mexican SMBs that automates who to contact first and what to say, while a human sends every message. Designed a rule-based 0-100 debtor priority score (days overdue, amount owed, unanswered messages, payment promises); an AI assistant (Gemini via Vercel AI SDK) that drafts reminders, suggests replies, detects payment promises, and writes a daily summary, with Zod-validated outputs (amounts never above the balance, dates always in the future) and rule-based fallbacks when the model or quota fails; and an in-app AI guide that uses tools to navigate the UI and highlight controls. Supabase (Postgres, Auth, row-level security, Realtime) isolates each user's data; AI endpoints stream NDJSON, carry per-user rate limits, and sit behind a provider-swappable model layer. WhatsApp Cloud API integration in progress, with the app submitted for Meta review in Oct 2026.

TypeScript, React, Hono, Vercel AI SDK, Gemini, Supabase, WhatsApp Cloud API

Hivara AI Collaboration - AI Engineer

Collaborated (June-Sept 2026) on Hivara, a production-grade WhatsApp conversational platform powered by a Python engine (FastAPI, Celery, Redis, RabbitMQ) with LLMs, vector RAG, and deep CRM/API integrations. Built a full local execution harness that ran the real engine and LLM while mocking external systems; redesigned a complex 100-node dealership flow (v29→v78) eliminating false terminations (**14/18→0/18**) and achieving 100% clean data capture; performed deep root-cause analysis of non-obvious engine behaviors (intent truncation, quick-match overrides, worker-level race conditions causing >1,200 CAS conflicts) that resulted in core engine fixes; executed large-scale E2E QA across 15 flows from 8 organizations; and delivered automation tooling, conversation scripts, and production-grade documentation.

Python, LangGraph, Celery, RAG

NexoTV - Streaming Platform

Designed a streaming backend in TypeScript focused on managing high user concurrency, Stripe-integrated subscriptions, and robust REST APIs.

TypeScript, PostgreSQL, Stripe, REST APIs

EDUCATION

Computer Engineering (Ingeniería Informática) Universidad Rafael Belloso Chacín 01/2025

Full Stack Development Henry 06/2024 - 09/2024

CERTIFICATIONS

TypeScript	Verified by Talently
Node.js	Verified by Talently
React JS	Verified by Talently

SKILLS

AI / LLM Engineering: LangGraph, LangChain, CrewAI, AutoGen, Semantic Kernel, LlamaIndex, Haystack, Smolagents, Mastra AI, Vercel AI SDK, OpenAI, Anthropic (Claude), Gemini, ElevenLabs (Text-to-Speech), Mistral, Cohere, Hugging Face, vLLM, Ollama, LiteLLM, Instructor, Outlines, Guardrails, MCP, Function Calling, Tool Use, Agentic Workflows

Python & Backend: Python, FastAPI, Django, Flask, Celery, RabbitMQ, Redis, WebSockets, ZeroMQ, Pydantic, SQLAlchemy, Prisma, asyncio, uvloop, httpx, Tenacity, Node.js, NestJS, Express.js, Fastify, Hono, Zod, GraphQL, Event-Driven Architecture, Go, Java/Spring Boot, Rust

RAG, Vector & Data: RAG, Advanced RAG, Hybrid Search, pgVector, Qdrant, Pinecone, Weaviate, Chroma, FAISS, Milvus, Elasticsearch, PostgreSQL, Supabase, MongoDB, MariaDB, SQL, Redis, MinIO, Data Modeling, Query Optimization, ETL Pipelines, LangSmith, Helicone, Phoenix, LlamaIndex, Haystack

ML / Evaluation & Observability: TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers, PEFT, LoRA, Evaluation (Ragas, DeepEval, Promptfoo), LangSmith, Helicone, Phoenix, Grafana, Prometheus, OpenTelemetry, Tracing, Prompt Engineering, Fine-tuning, Embeddings

Cloud, DevOps & Tools: AWS (Bedrock, SageMaker, Lambda), Google Cloud (Vertex AI), Render, Kubernetes, Docker, CI/CD, Microservices, Terraform, Linux, Git, GitHub, GitLab

Enterprise Integrations: Meta Business API, WhatsApp Business API, Instagram Graph API, Facebook API, EspoCRM, Twilio, Mailchimp API, ElevenLabs API, NFC Hardware Integration, Webhooks, API Integration

Frontend: React, Next.js, Vite, TypeScript, Tailwind CSS, Flutter **Languages:** Spanish (Native), English (B2 - Upper-intermediate)